

Izmir Institute of Technology

Electrical and Electronics Eng. Dep.

EE 471 Modern Software Development Practices and Technologies In-class exercise Week #10

Thursday 07/05/2026

Task 1: Read through the HuggingFace LLM course doc and implement: them in Python.

[Transformers and sentiment analysis:](#)

- Apply a sentiment analysis for the following two sentences: **“I've been waiting for a EE471 course my whole life.”** and **“I hate EE471 course”**
- Zero-shot classification for: **“Berkshire keeps their cash reserves at an extremely high level.”**
- Complete the following sentence **“If I continue to successfully complete all in-class exercises in EE471 course,”** with a maximum length of 35 words, return two alternative sentences.
- Use transformers to fill the gap in the following sentence **“To understand generative AI, one must study [...] well.”**
- Use named entity recognition to extract the name of the subject, which company do they work, and where do they live from the following sentence **“I am Nate, a research assistant in Izmir Institute of Technology, and currently living and working in beautiful city İzmir in Türkiye.”**
- Validate the extracted person, organization, and city info above via question answering.
- Summarize the following text: **“The 2008 Global Financial Crisis stands as the most severe economic collapse of the 21st century, often compared to the Great Depression of the 1930s. Triggered by the bursting of the United States housing bubble, its effects rippled across the globe, leading to the collapse of major financial institutions and a deep international recession. The crisis began with the subprime mortgage market. In the early 2000s, low interest rates and a push for homeownership led banks to issue high-risk loans to borrowers with poor credit.”**
- Translate the following sentence: **“The 2008 Global Financial Crisis stands as the most severe economic collapse of the 21st century, often compared to the Great Depression.”**
- Use Google's /vit-base-patch16-224 for a basic image classification task.
- Use openAI's whisper-large-v3 model for automatic speech recognition.

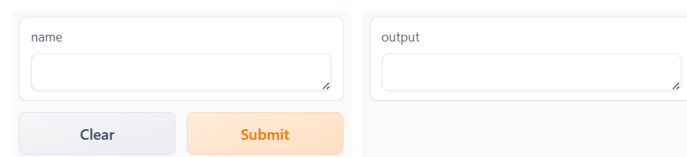
Deliverables (Deadline: Until the end of the lecture): Link for your github repo (latest push).

Task 2: Read through the HuggingFace LLM course doc ([2nd chapter](#)) and implement the following tasks in Python:

- In OOP manner write a python module capable of each of the tasks in Task 1:

Sentiment analysis	Question Answering
Zero-shot classification	Text Summarization
Text generation (from an incomplete sentence)	Text Translation
Mask filling	Image classification
Named entity Recognition	Automatic speech recognition

- Use this software module and Gradio to develop an interactive demo for each of the tasks with a Gradio-based front end, e.g.:



The image shows a Gradio interface with two main components. On the left, there is a text input field labeled 'name' with a small icon in the bottom right corner. Below this field are two buttons: a grey 'Clear' button and an orange 'Submit' button. On the right, there is a text output field labeled 'output' with a small icon in the bottom right corner.

Deliverables (Deadline: until midnight):

- 1) The link for your public github repository
- 2) A screen recording video demonstrating each of the tasks: **Task 1 & Task 2, with the explanation of code blocks** similar to the previous week's exercise.